Process Management

DR. MEETA BHATIA

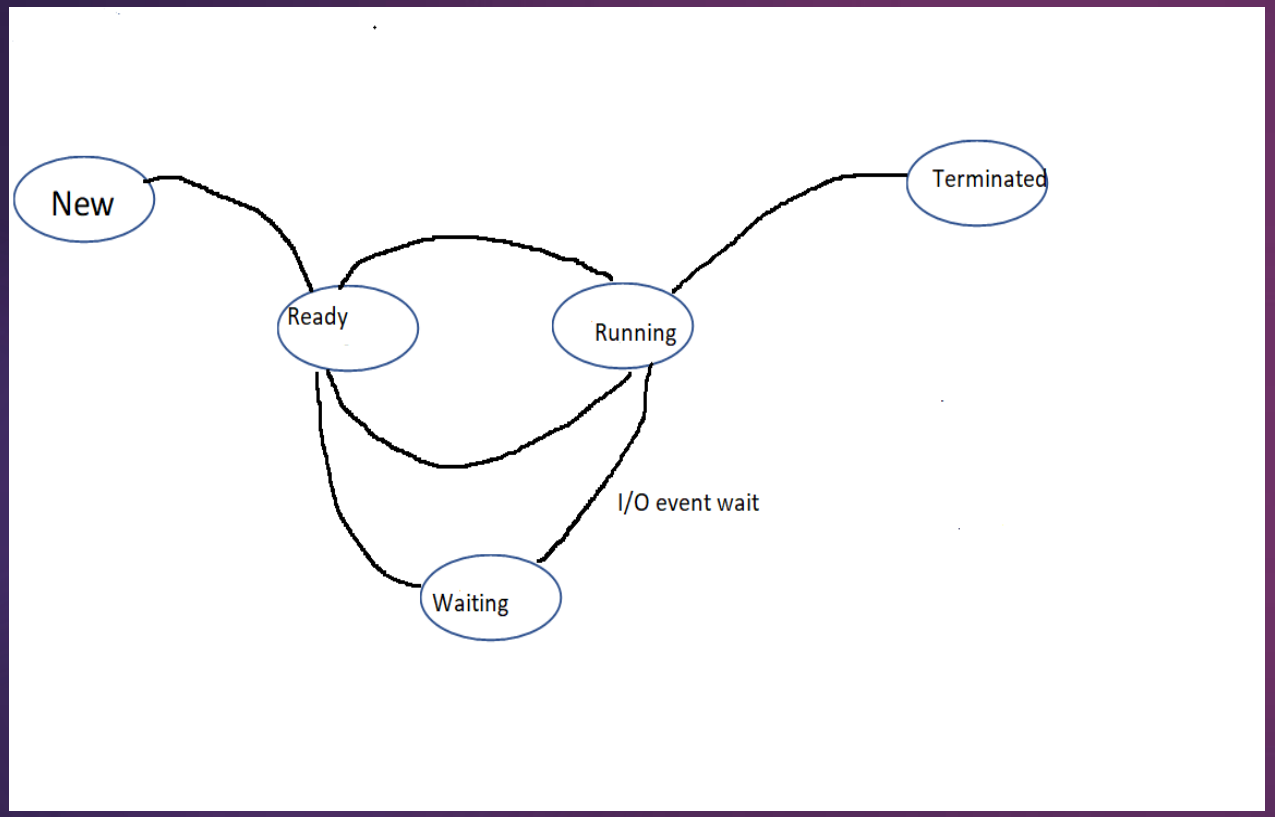
Program and Process.

A program is a set of instructions , written by a user or a programmer and stored somewhere. It means that a program is a passive entity and continues to exist at a place. On the other hand, when a program is ready for execution, it becomes active and know as a process. It can be thought of as a program in execution. It is the unit of work in most systems.

OS as Process Manager

- Scheduling processes and threads on the CPU.
- Create and destroy Processes.
- Suspending and resuming processes.
- Allocating Hardware resources to the process
- Providing a means of communication or signal among processes.

Process State :



Process State:

- New: Process is being created.
- Ready : The process is waiting to be assigned to a processor.
- Running : Instruction are being executed
- Waiting : Process is waiting for some event to occur.
- Terminated : The process has finished execution.

Process Control Block

Process Id
Process State
Program Counter
Registers
Pointers
Memory Limits
List of Open Files
⋮

Scheduling

Scheduling refers to a set of policies and mechanism supported by OS that controls the order in which the work to be done is completed.

Scheduling Criteria

- **CPU Utilization** : Key idea is that if the CPU is busy all the time, the utilization factor of all the components of the system will also be high.
- **Throughput** : It refers to the amount of work completed in a unit of time. One way to express throughput is by means of the number of user's jobs executed in a unit of time. Higher number represents more work is being done by the system.
- **Turnaround time** : Time that elapses from the moment a program or a job is submitted until it is completed by the system. It is the time spent in the system and may be expressed as a sum of the job service time and waiting time.

Turnaround time=∑(Period spending waiting to get into memory, waiting in the ready queue, Executing on CPU & doing I/O)

Cont……

- **Waiting Time** : Time that a process or a job spends waiting for resource allocation due to contentions with others in a multi programming system.

$W(x) = T(x) - x$

X : Service time

W(x): Waiting time of the job requiring x units of service.

T(x) : Job's turnaround time.

- **Response time** : This is the time from submission of a request until the 1st response is produced. It is the amount of time , system takes to start responding.

Purpose of Scheduling

Maximize

- CPU Utilization
- Throughput,

- Minimize
  - turnaround time,
  - Waiting time, and
  - Response time.

Scheduler :

Is an OS module that selects next job to be admitted for execution.

- Long Term Scheduler
- Short Term Scheduler
- Mid Term Scheduler

Scheduling Algorithms

- FCFS – First Come First Serve

with this scheme, the process that requests the CPU first is allocated the CPU first.

Implementation is simple.

Execution results little overhead.

Implemented through FIFO queue. When the process enters the ready queue, its PCB is linked onto the tail of the queue.

When CPU is free, it is allocated to the process at the head of the queue. The running process is then removed from the queue.

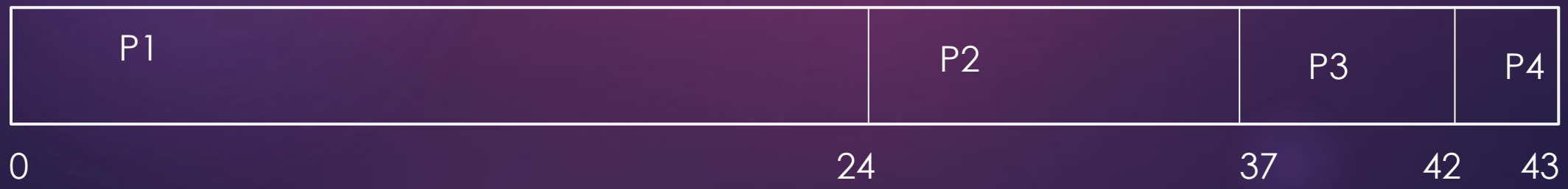
Average waiting time under the FCFS policy is quite long.

FCFS example :

Example :

Process	Burst Time	TAT	WT
P1	24	24	0
P2	13	37	24
P3	5	42	37
P4	1	43	42
Avg		146/4=36.5	103/4=25.75

GANTT Chart



FCFS is non pre-emptive Scheduling algorithm. Once the CPU has been allocated to a process, that process keeps the CPU until it release the CPU, either by terminating or by I/O request.

SJF : Shortest Job First

The next scheduling entity (process) is selected on the basis of shortest execution time.

When CPU is available, it is assigned to the process that has smallest next CPU bust.

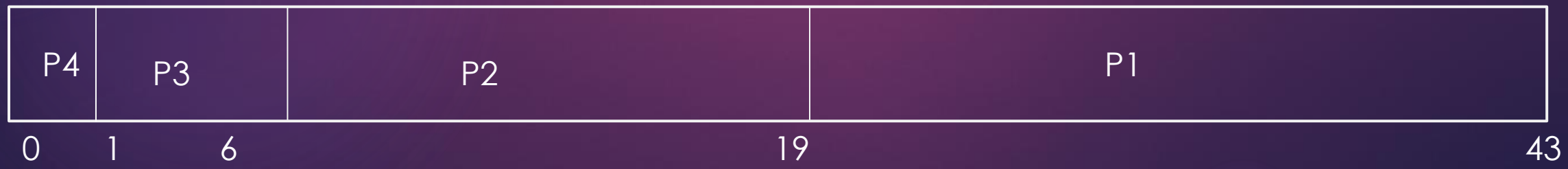
If 2 processes has same burst time, FCFS is used to select one of them

This algorithm is non pre-emptive.

SJF example

Process	BT
P1	24
P2	13
P3	5
P4	1

Process	BT	TAT	WT
P4	1	1	0
P3	5	6	1
P2	13	19	6
P1	24	43	19
Avg			69/4=17.25	26/4=6.5



Round Robin

- Processor time is divided into slice or quanta, which is allocated to requesting processes.
- No process can run for more than one time slice, if there are processes waiting in the ready queue.
- Process leaves queue, if the task is over.
- After exhausting its time slice, if any process needs more time, it is placed at the end of the queue.
- RR scheduling achieve equitable sharing of resources. Short processes may be executed within a single time quantum showing a good response time.

Advantages :

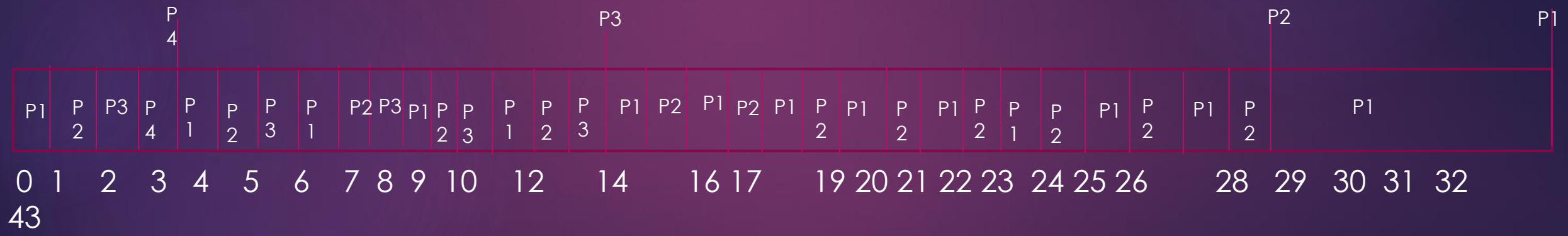
- Fair scheduling Algorithm.
- One of the best known scheduling disciplines for achieving good and relatively evenly distributed response time.

Size of Time Slice and performance:

- Reduction in Time slice gives better response time but lots of overhead due to frequent timer interrupts and process switches.
- Too long time slice reduces overhead but increases response time too.

Example RR

Process	Burst Time	TAT	WT
P1	24	43	19
P2	13	32	19
P3	5	16	11
P4	1	04	03
Avg		95/4=23.75	52/4=13



Comparison among these 3 algorithms:

	Avg TAT	Avg WT
FCFS:	36.50	25.75
SJF	17.25	06.50
RR	23.75	13.00